

Fused Filament Fabrication

Standard Operating Procedure

Equipment Location: PawPrint Studio

Ingram Hall Makerspace
Ingram School of Engineering
College of Science and Engineering



Policy Reminders

- Reserving a 3D Printer in FOM is not only required, it is impossible to print anything without doing so
- The following trainings are required for using the Prusa CoreOne or Prusa Mini printers
 - Fused Filament Fabrication Policy and Procedures Quiz (FOM)
 - LinkedIn Learning FFF Certificate
- In addition to the above trainings, the following trainings are also required for using the PrusaXL printers:
 - Multi-Material FFF Policy and Procedures Quiz (FOM)

Policy Reminder (cont.)

- Unless a printer is labeled “BYOF” the only allowable filament is PLA supplied by the Makerspace
- Printers labeled “BYOF” may not use any Makerspace filament and the user supplies their own
 - Any filament is allowed, provided it matches the capabilities of the printer
 - Ask a Makerspace staff member for questions regarding material compatibility
- There is a 30-minute minimum charge for all 3D Printer usage
- Users are permitted to check out multiple printers simultaneously, with no set maximum limit.
- Avoid touching moving parts while printing
- Be careful of the printer's hot end

SOP Outline

1. [Using Orca Slicer](#)
2. [Accessing the Printer WebUI](#)
3. [Starting Your Print Using OctoPrint](#)
4. [Appendix](#)

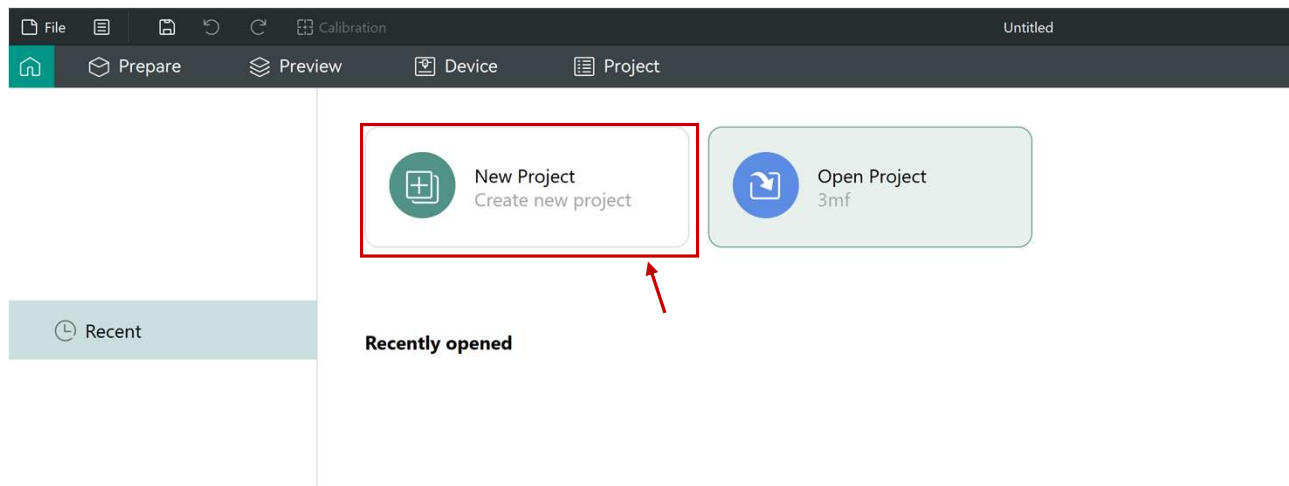
Using Orca Slicer

- Orca Slicer is the recommended slicing program for all Prusa printers managed by the Makerspace
 - Printers and profiles are pre-configured on Makerspace computers
- When using orca slicer make sure all your printer requirements are correct.
 - Specifically, ensure the correct printer and filament type are selected
- To start first open the application. The icon should look like this:



Using Orca Slicer

- Once open, you should be at the homepage.
- Click “New Project” to begin.

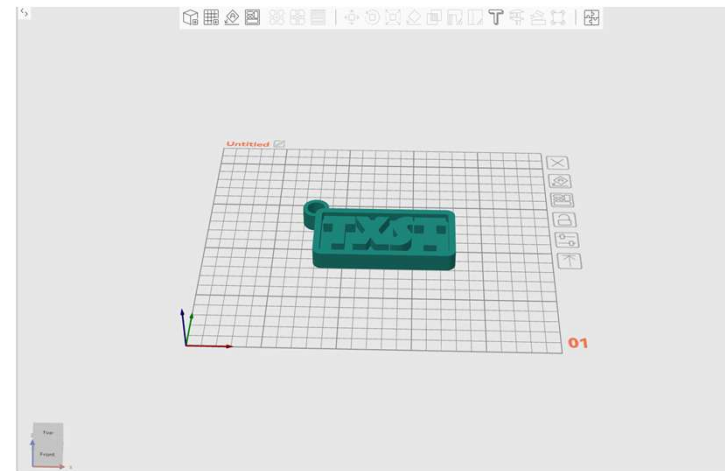
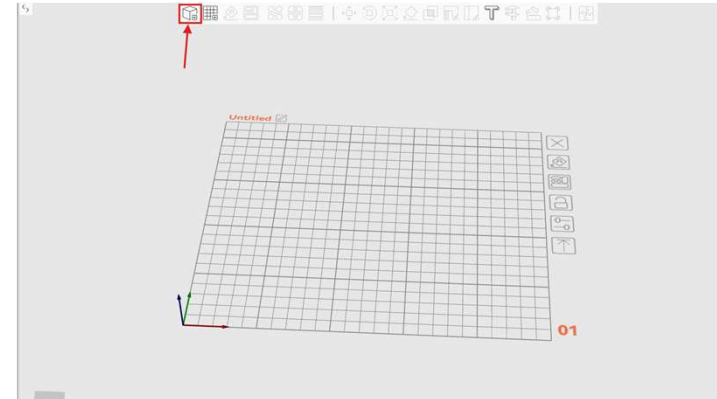


Importing files to Orca

- To import files to Orca click the "add" icon at the top. It should look like a cube.



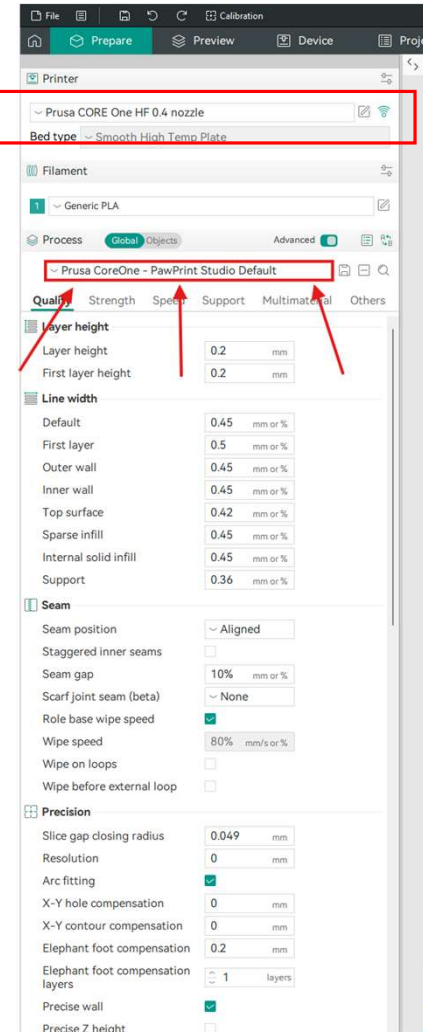
- Once clicked, select your file (.stl or .3mf) and upload.
- Your selected file should now appear on the build plate.



Using Orca Slicer

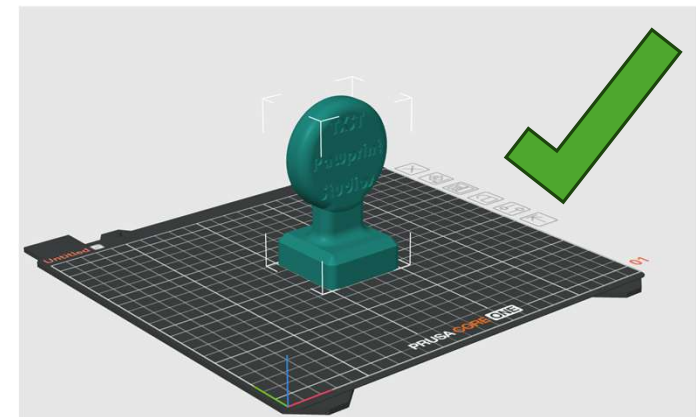
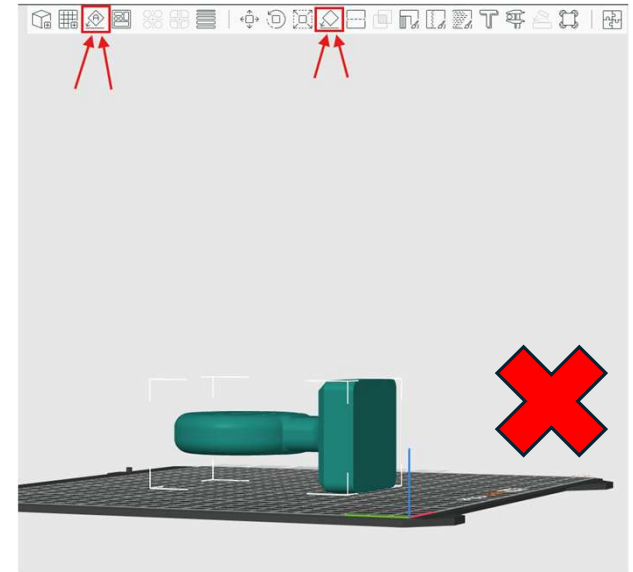
- You can now prepare your print
- On the lefthand side of the prepare page are all your printer settings.
 - Ensure the correct printer is selected
 - Prusa CoreOne = Prusa CORE One HF 0.4 nozzle
 - Prusa Mini = Prusa MINIIS 0.4 nozzle
 - In general, most prints will be successful by selecting "***PawPrint Studio Default***"

Printer Selection



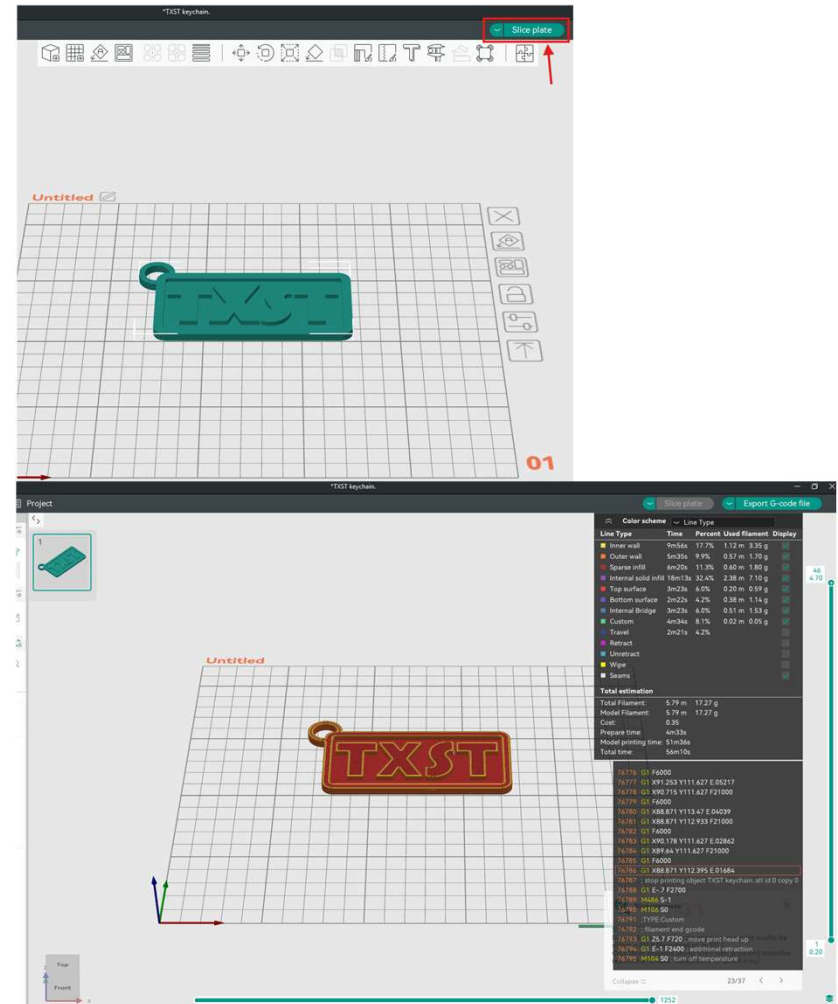
Print orientation

- Once your object is imported it might be in the wrong orientation and will be unable to print properly.
 - You will typically receive an error regarding hanging regions
- Ensure that you click either "Auto orient" or "lay on face" to properly orient your object.
- Your object should now be in the correct orientation to print.



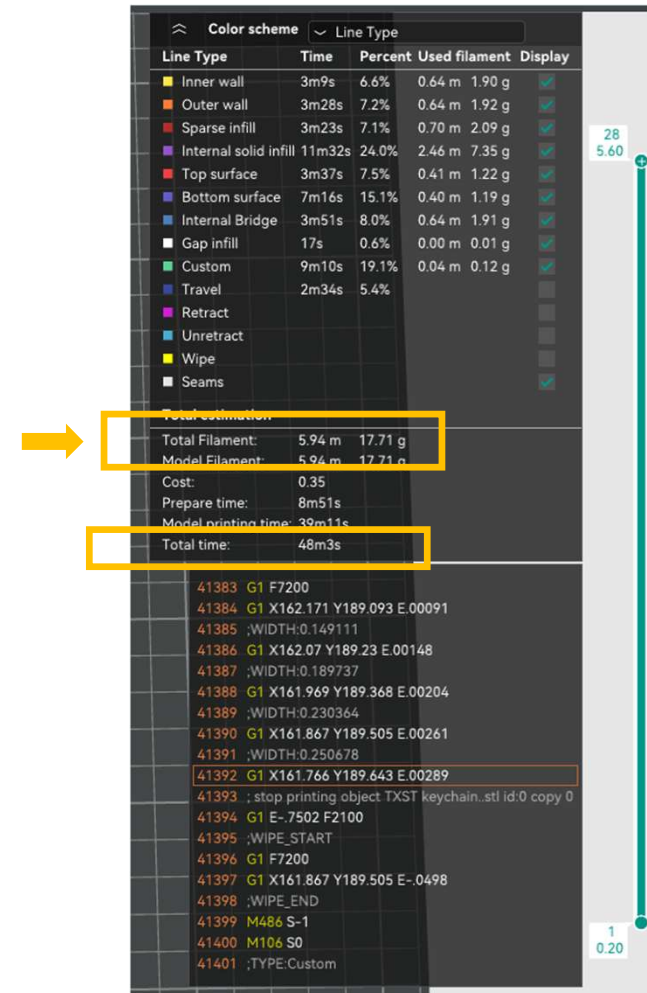
Slicing an object

- Ensure that the appropriate printer is selected
 - Prusa CoreOne
 - Prusa Mini
 - Prusa XL (requires additional training)
- Ensure that the correct slicer settings are applied (profile)
- Ensure that the correct material is selected (PLA)
- Then to slice an object, click the "Slice Plate" button
- You should now have a sliced object ready for export



Slicing information

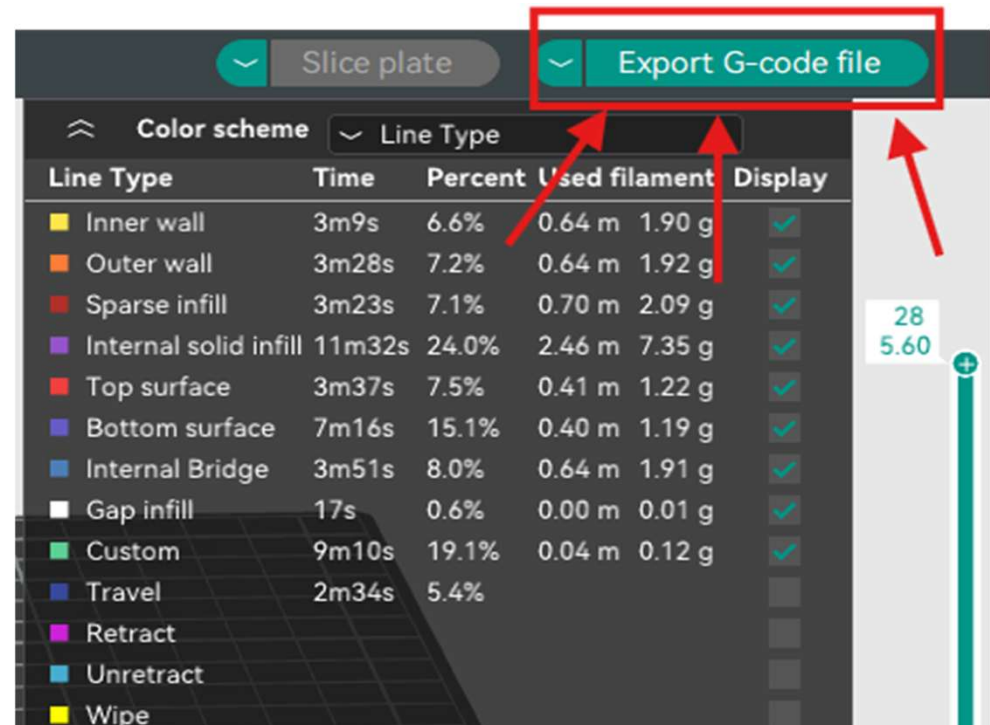
- On the right-hand side of the screen a menu showing all the slicers information will appear.
 - Take note of:
 - Estimated total filament**
 - Total time**
 - You will need both numbers later
- This menu will break down all the 3d printers' movements and its time completing a section.
- If you would like to check each layer. There are sliders on the right and bottom of the screen that you can adjust.



Line Type	Time	Percent	Used filament	Display
Inner wall	3m9s	6.6%	0.64 m 1.90 g	✓
Outer wall	3m28s	7.2%	0.64 m 1.92 g	✓
Sparse infill	3m23s	7.1%	0.70 m 2.09 g	✓
Internal solid infill	11m32s	24.0%	2.46 m 7.35 g	✓
Top surface	3m37s	7.5%	0.41 m 1.22 g	✓
Bottom surface	7m16s	15.1%	0.40 m 1.19 g	✓
Internal Bridge	3m51s	8.0%	0.64 m 1.91 g	✓
Gap infill	17s	0.6%	0.00 m 0.01 g	✓
Custom	9m10s	19.1%	0.04 m 0.12 g	✓
Travel	2m34s	5.4%		
Retract				
Unretract				
Wipe				
Seams				✓
Total estimation:				
Total Filament:	5.94 m	17.71 g		
Model Filament:	5.94 m	17.71 g		
Cost:	0.35			
Prepare time:	8m51s			
Model printing time:	39m11s			
Total time:	48m3s			
41383 G1 F7200				
41384 G1 X162.171 Y189.093 E.00091				
41385 ;WIDTH:0.149111				
41386 G1 X162.07 Y189.23 E.00148				
41387 ;WIDTH:0.189737				
41388 G1 X161.969 Y189.368 E.00204				
41389 ;WIDTH:0.230364				
41390 G1 X161.867 Y189.505 E.00261				
41391 ;WIDTH:0.250678				
41392 G1 X161.766 Y189.643 E.00289				
41393 ; stop printing object TXST keychain.stl id:0 copy 0				
41394 G1 E-7502 F2100				
41395 ;WIPE_START				
41396 G1 F7200				
41397 G1 X161.867 Y189.505 E-.0498				
41398 ;WIPE_END				
41399 M486 S-1				
41400 M106 S0				
41401 ;TYPE:Custom				

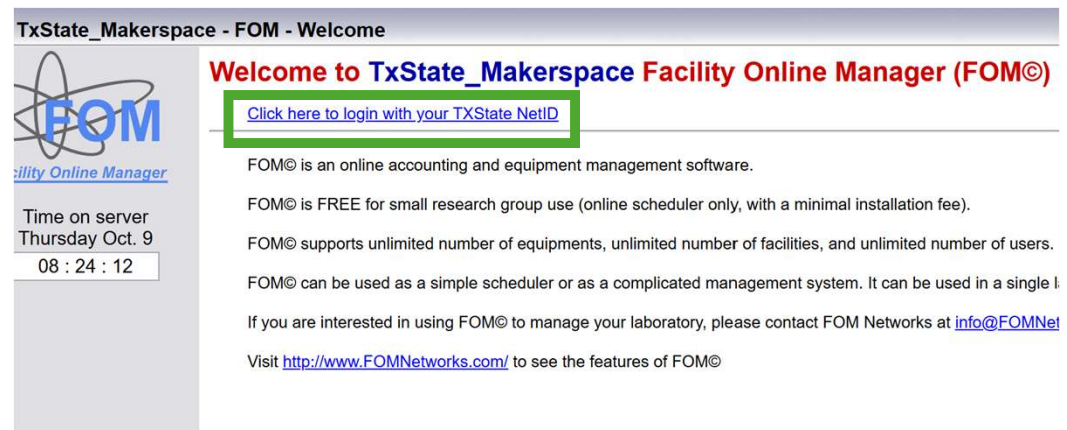
Exporting your G-code file

- Click "Export G-code file" and save to a known location for uploading to the printer.
 - You may need to click the drop-down arrow if this option is not immediately available



Sign into FOM

- After slicing, sign into your FOM account
- Click the ***Click here to login with your TXState NetID*** link
- Sign in using your TXST NetID and password



Checking out a printer

- Decide on an available printer to checkout
- To add a new printer to your profile, click:
 - (Dashboard View) **Request Access to New Resources** button at the top of the page
 - (Classic View) the drop down + icon next to Ingram makerspace under Available Resources
- If your training prerequisites are current, no new training will be required.

Dashboard View

Request Access to New Resource

Classic View

Available Resources (Click to request training or usage approval) [Hide this section](#)

Select or type to search

+ Ingram Makerspace

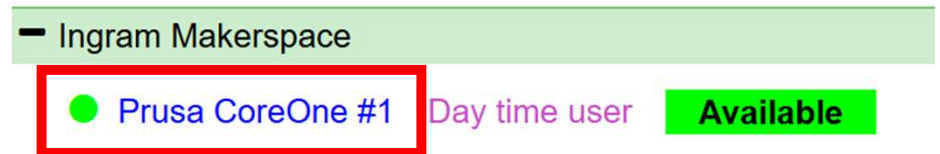
Accessing the Equipment Calendar

- In Dashboard View
 - Click the calendar icon
- In Classic View
 - Click the Equipment Name

Dashboard View



Classic View



Reserving Equipment Time

- Click on the starting time you wish to reserve your printer
- A pop-up will appear asking for details of your reservation
 - Financial Account: This is your course #, Research Project, or Personal Project name
 - Reservation: Choose a start time and end time.
 - *If you wish to print immediately, choose the closest start time to the current time.*
 - The total time should be the estimate time value from Orca Slicer + 45 minutes to allow for printer warm-up and leveling, and clean-up
 - Estimated Cost: This is auto-calculated based on reservation time. This is calculated for internal rates, regardless of which account you are using.
 - Usage Comments: Optional text field
 - Consumables: Enter the amount of filament used (in grams) as shown in Orca Slicer
 - For BYOF printers, this option will not appear

Prusa CoreOne #1 Hide description Available Day time user
 The CORE One is a fully enclosed CoreXY 3D printer with active temperature control designed for top print quality and speed. Easy to set up and made to last, this rugged machine is perfect for beginners and professionals alike.

Oct 2025 <Prev Today Next>

Mon 10/6	Tue 10/7	Wed 10/8	Thu 10/9	Fri 10/10	Sat 10/11	Sun 10/12
7:30						
8:00	8:00	8:00	8:00	8:00	8:00	
8:30	8:30	8:30	8:30	8:30	8:30	
9:00	9:00	9:00	9:00	9:00	9:00	
9:30	9:30	9:30	9:30	9:30	9:30	
10:00	10:00	10:00	10:00	10:00	10:00	
10:30	10:30		10:30	10:30	10:30	
11:00	11:00		11:00	11:00	11:00	
11:30	11:30	11:30	11:30	11:30	11:30	
12:00	12:00	12:00	12:00	12:00	12:00	
12:30	12:30	12:30	12:30	12:30	12:30	
13:00	13:00	13:00	13:00	13:00	13:00	
13:30	13:30	13:30	13:30	13:30	13:30	
14:00	14:00	14:00	14:00	14:00	14:00	
14:30	14:30	14:30	14:30	14:30	14:30	

Reserving Equipment Time

New reservation ×

Resource: Prusa CoreOne #1

User: You

Fee structure: Internal

Access Level: Day time user

Financial account: ProjectDescription ▾

Reservation: 10/9 9:30 ▾ - 10/9 15:00 ▾

Estimated cost: 3.3

Usage Comment:

Enter starting time

+ [Send email/meeting invite](#)

Consumables: PLA Filament 100 gram

Enter estimated filament used from slicer

Click when finished with details

Reserve

Close

Reserving Equipment Time

- Printers are only powered on when equipment is checked out
 - Your reservation is cancelled if you do not check out the equipment within 30 minutes of your reservation time.
- Printers are powered off when equipment is checked back in
 - Manual log-off should occur after your print is finished, however:
 - Auto-log off (and power-off) occurs four hours after reserved end-time
 - If your print extends past the reserved end-time, the next reserved user may log you off, effectively ending your print
- **It is important to set your reservation time correctly. You may extend your reservation time as needed if no reservations are after you.**

Logging onto Printer

- After reserving equipment, click on your reservation in calendar view. You will see a pop-up of your reservation details. Click **Logon** to check-out the printer



Current reservation ×

Comment by previous user:
//Consumables at logoff: None//

Resource: Prusa CoreOne #1

User: [REDACTED]

Fee structure: Internal

Access Level: Day time user

Financial account: ProjectDescription ▾

Reservation: 10/9 9:30 - 10/9 15:00

Usage Comment:
//Consumables when reserve: PLA Filament 100.00 gram at 0.02/gram//

Consumables: PLA Filament gram

Record ID 14 created at 10/9 9:00, last updated at 10/9 9:00

Logon Close

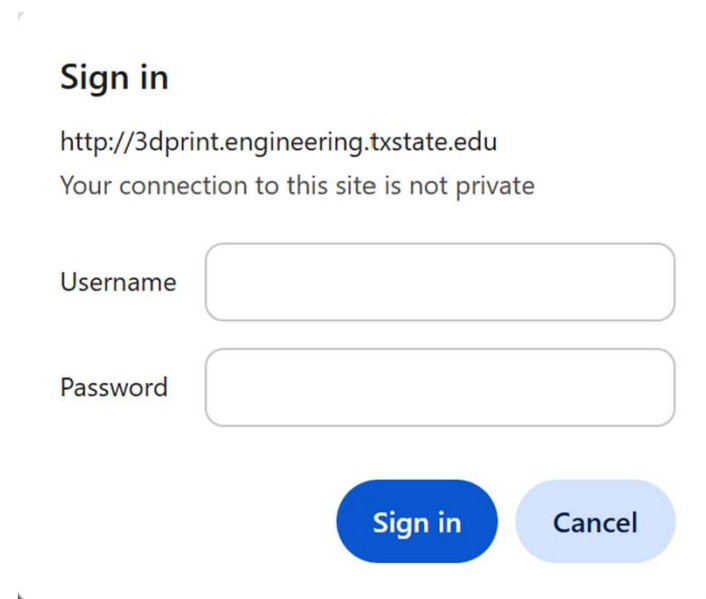
Accessing the Printer's Interface

- Navigate to:

3dprint.engineering.txstate.edu

(only accessible when connected to the TXST network)

- Depending on which computer you are using, a login prompt may appear. Use your TXST NetID and password if prompted.
 - If accessing from a personal computer and your login does not work, you may need to insert "TXSTATE\" before your username
 - Example: TXSTATE\boko123



Sign in

http://3dprint.engineering.txstate.edu

Your connection to this site is not private

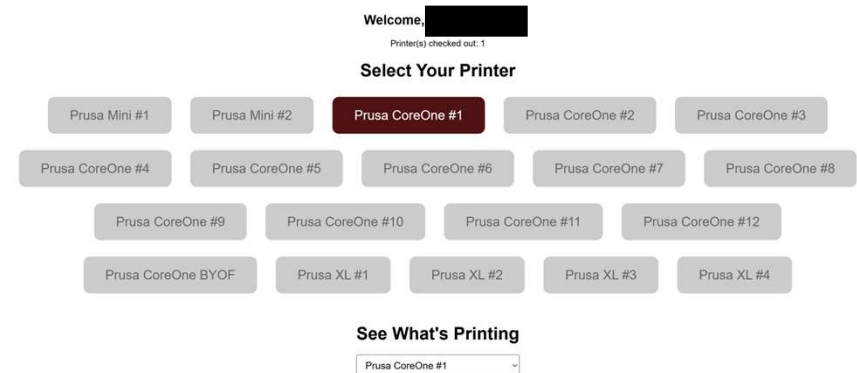
Username

Password

Sign in **Cancel**

Accessing the Printer's Interface

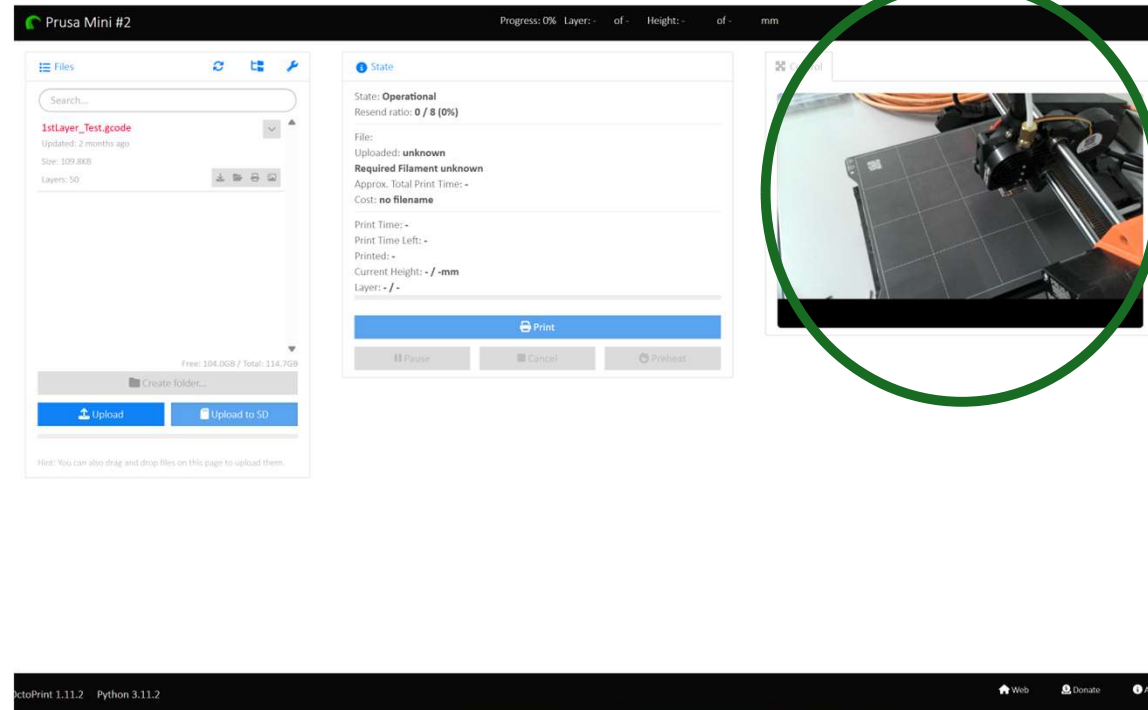
- The printer(s) you currently have checked out will be available as clickable buttons
 - Click on your checked out printer to proceed
- Printers not checked out to you will be disabled.
- All printers that are checked out will have their camera feeds shown



Using Octoprint

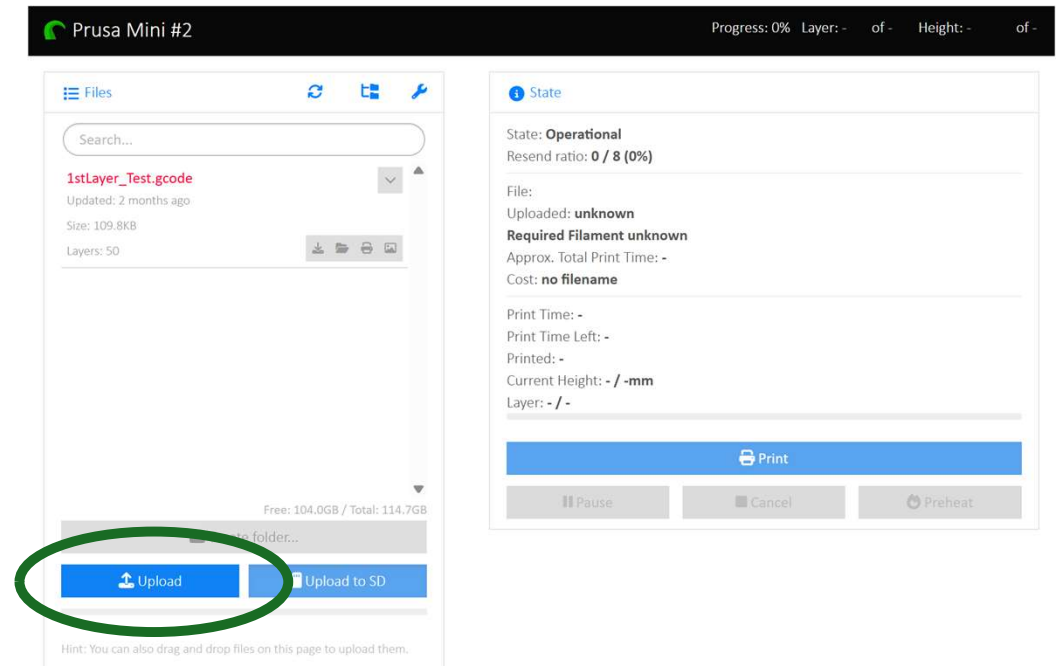
- Once redirected to your printer's Octoprint instance, ensure:
 - The build plate is clear
 - There is enough filament for your print
 - You will need to physically check the printer
 - Ask an MST if the spool is low

Build Plate Clear



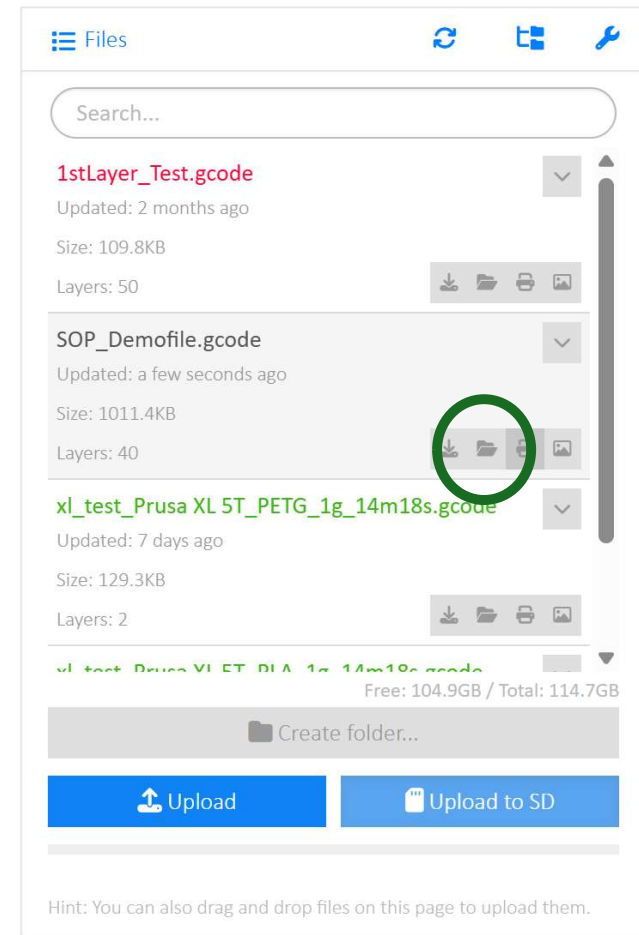
Uploading your Gcode

- Click the **upload** button on the left-hand side of the screen and select your file.
 - You may also drag and drop files to upload
- Files delete weekly, so do not rely on printer SD cards for personal storage.



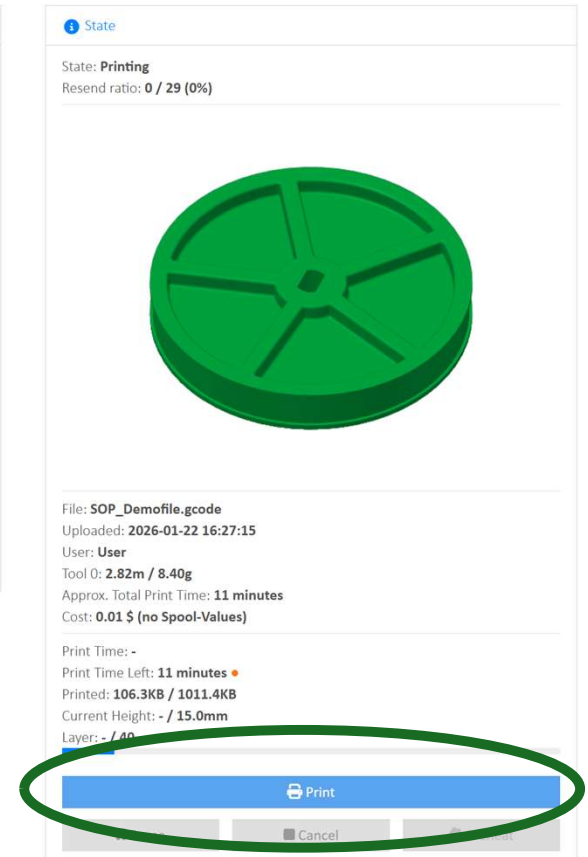
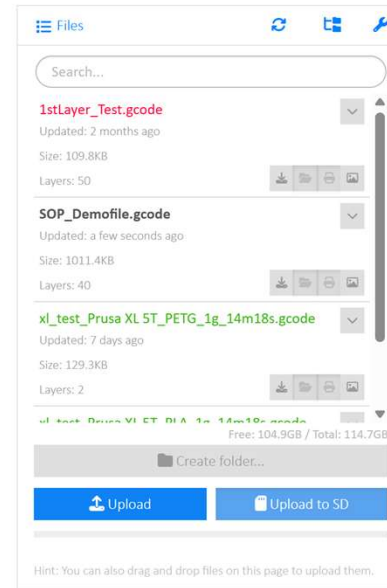
Starting your print

- Click the folder icon next to your file to load your gcode for printing



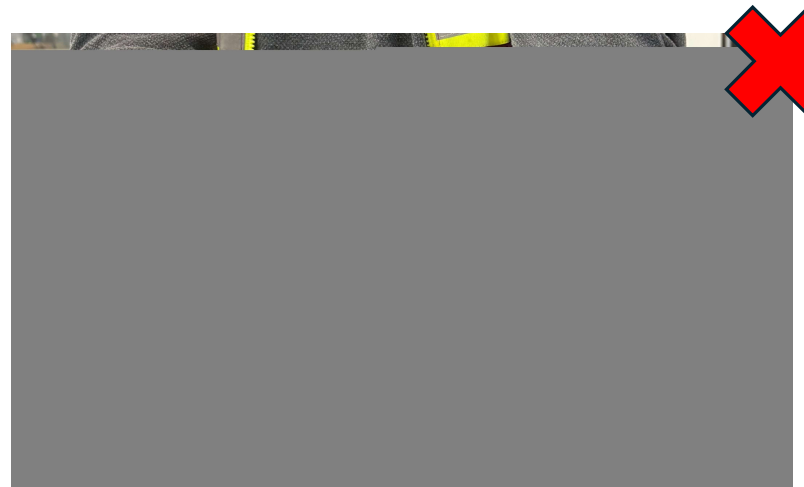
Starting your print

- Your model should be loaded under the “State” box
- Click **print** to start printing
 - Most print failures occur within the first 30 minutes. You must monitor your print during this time.



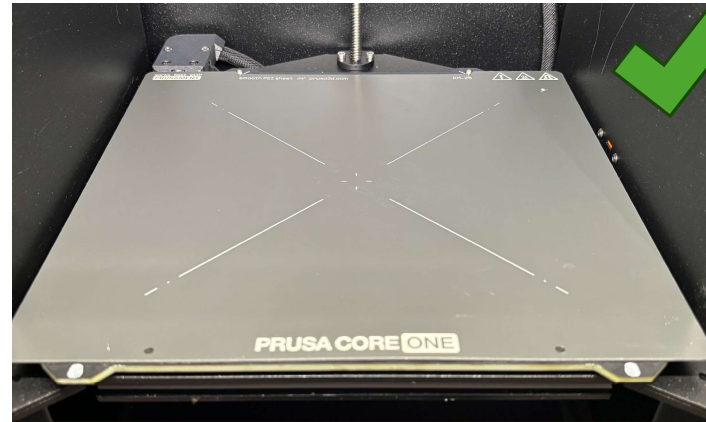
Print Finished

- When your print is finished, remove your part from the build plate
- **Do not use any tools on the print bed.**
- You may slightly bend the build plate to decrease part adhesion to remove your part. **Do not excessively bend the build plate.**



Print Finished

- Ensure the build plate is free of any debris or run-up filament and return it to the printer in the correct orientation



Print Finished

- Return to the FOM page
- Select your reservation
- In the pop-up, adjust the filament used if different from slicer estimates
 - If using a BYOF printer this should not appear
- Check back in your printer by clicking the **Logoff** Button

Resource is being used

Resource: Prusa CoreOne #1
User: [REDACTED]
Fee structure: Internal
Access Level: Day time user
Financial account: ProjectDescription
Reservation: 10/9 9:30 - 10/9 15:00 (Select a later time to extend your session)
Actual usage: 10/9 9:54 - 10/9 10:07
Resource status report: ☒ OK ☐ Something wrong (details in comments)
Usage Comment:
//Consumables when reserve: PLA Filament 100.00 gram at 0.02/gram// //Consumables at logon: PLA Filament 100.00 gram at 0.02/gram//
Consumables: PLA Filament 100 gram
Record ID 14 created at 10/9 9:00, last updated at 10/9 9:54
Logoff Extend Close

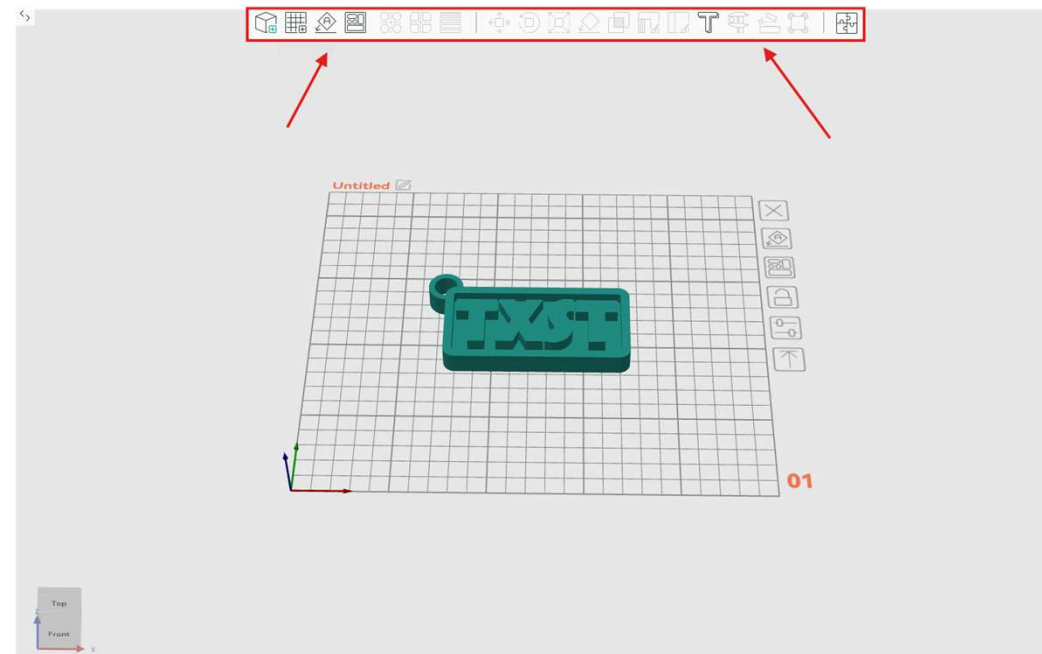
Adjust as needed

Appendix

Additional information on Orca Slicer

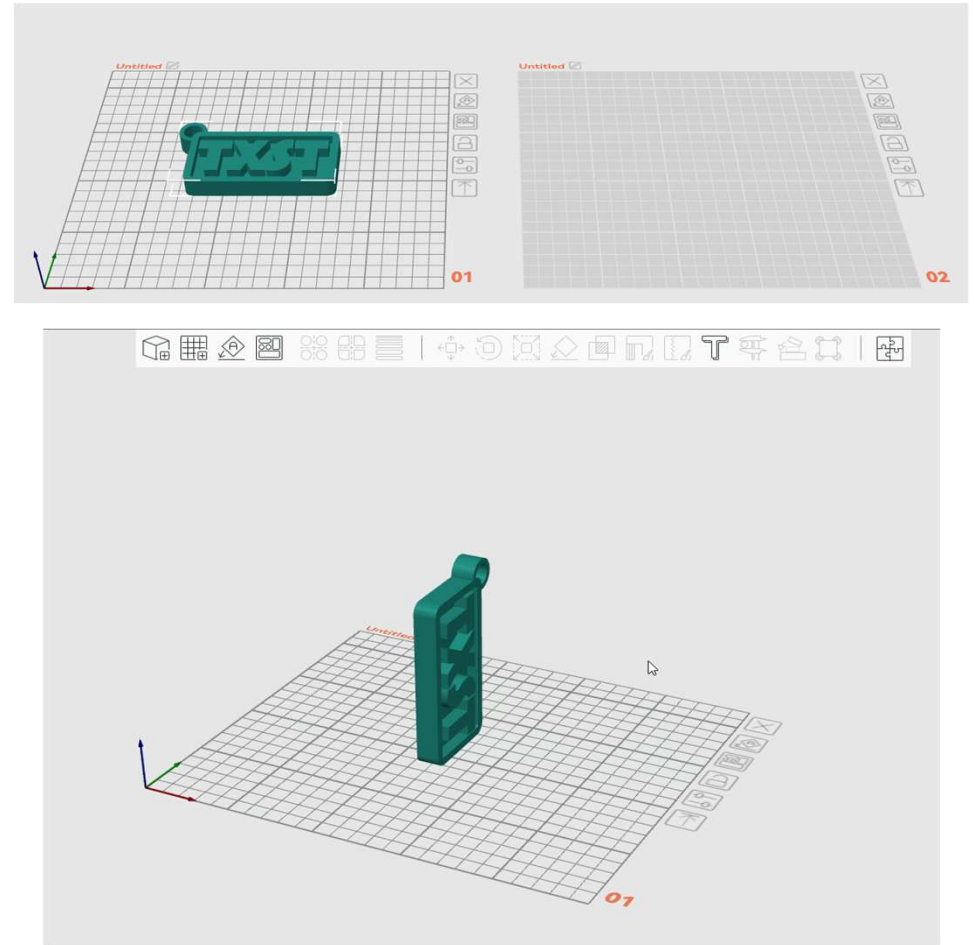
Orca Slicer Toolbar

- The tool bar is located at the top of the screen in the middle.
- From left to right the tools include; "Add", "Add Plate", "Auto Orient", "Arrange", "Split to objects", "Split to parts", "Variable layer height", "Move", "rotate", "Scale", "Lay on face", "Mesh Boolean", "Support/Seam painting", "Emboss", "Measure", "Assemble", "Brim ears", and "Assembly View".



Toolbar specifics

- Add:
 - Lets you upload your stl file onto the build plate in Orca
- Add plate:
 - When working on multiple stl files you can add additional plates for convince
- Auto Orient:
 - Orca will automatically orient your object to what it thinks would be the best surface to print on. If it's not to your liking, you can change it manually
- Arrange all objects:
 - Will auto arrange multiple objects on single plate for best printing

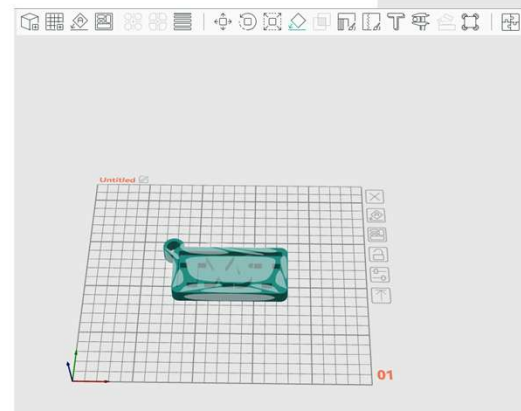
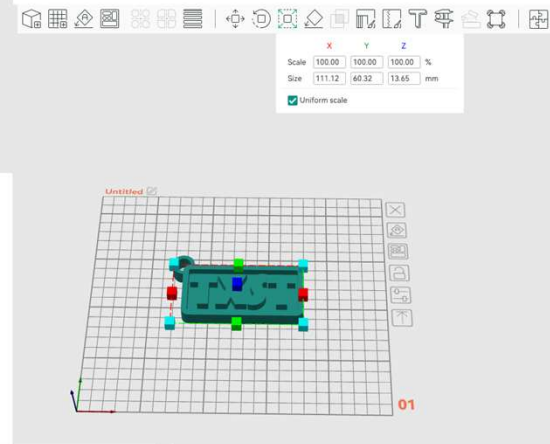
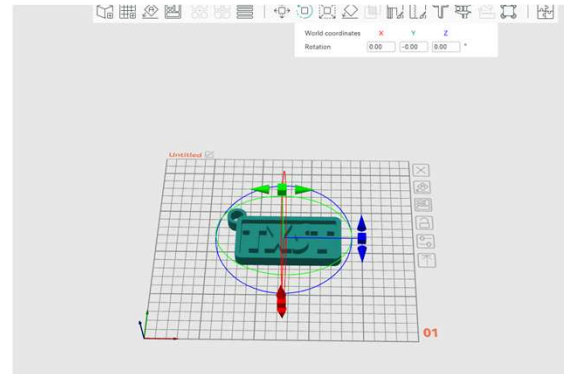


Toolbar specifics

- Split to Objects:
 - Will split selected object into multiple objects
- Splits to Parts:
 - Will split selected object into multiple parts
- Variable layer height:
 - Can change the layer height of your selected object
- Move:
 - Can move any selected object

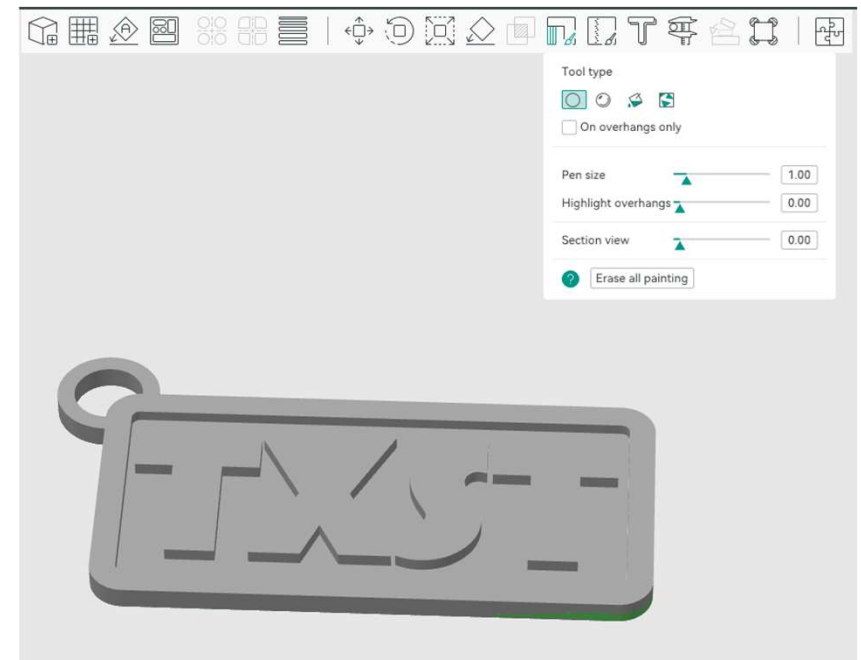
Toolbar specifics

- Rotate:
 - Will rotate your selected object
- Scale:
 - Will scale your selected object
- Lay on face:
 - Will display multiple different faces that you can lay your object on
- Mesh Boolean:
 - Will physically combine two separate parts into one.



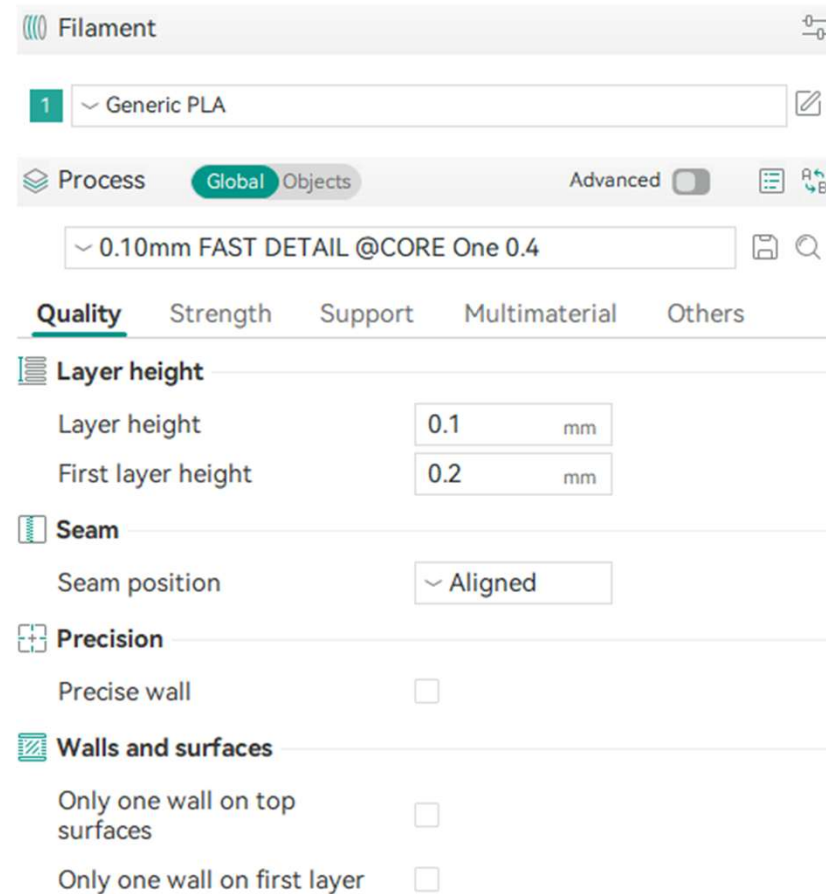
Toolbar Specifics

- Support Painting:
 - Allows you to choose exactly where you do or do not want supports
- Seam Painting:
 - Removes unwanted seams on prints
- Emboss
 - Allows you to add text to your selected object



Changing the settings

- On the lefthand side there are multiple settings when printing such as: "Quality", "Strength", "Support", "Multimaterial", "Filament", and "Other".



The screenshot shows the Cura software interface for configuring a filament profile. The top section is titled "Filament" and includes a dropdown menu for "Generic PLA". Below this, the "Process" tab is selected, showing a dropdown for "0.10mm FAST DETAIL @CORE One 0.4". The "Quality" sub-tab is active, displaying settings for "Layer height" (0.1 mm), "First layer height" (0.2 mm), "Seam" (Aligned), "Precision" (Precise wall), and "Walls and surfaces" (Only one wall on top surfaces, Only one wall on first layer).

Filament

1 Generic PLA

Process Global Objects Advanced

0.10mm FAST DETAIL @CORE One 0.4

Quality Strength Support Multimaterial Others

Layer height

Layer height 0.1 mm

First layer height 0.2 mm

Seam

Seam position ~ Aligned

Precision

Precise wall ☐

Walls and surfaces

Only one wall on top surfaces ☐

Only one wall on first layer ☐

Quality and Strength

- Quality:
 - In quality you can change the Layer height, the Seam, Precision, and lastly Walls and surfaces. You can hover your cursor over the setting and read more about it
- Strength:
 - In Strength you can change the Walls, Top and Bottom, and Infill. You can hover your cursor over the setting and read more about it

Quality	Strength	Support	Multimaterial	Others
Layer height				
Layer height		0.1	mm	
First layer height		0.2	mm	
Seam				
Seam position		~ Aligned		
Precision				
Precise wall		☐		
Walls and surfaces				
Only one wall on top surfaces		☐		
Only one wall on first layer		☐		
Quality	Strength	Support	Multimaterial	Others
Walls				
Wall loops		3		
Top/bottom shells				
Top shell layers		8	layers	
Top shell thickness		0.7	mm	
Top surface pattern		Monotonic ...		
Bottom shell layers		7	layers	
Bottom shell thickness		0.5	mm	
Bottom surface pattern		Monotonic		
Infill				
Sparse infill density		15	%	
Sparse infill pattern		Cross Hatch		
Internal solid infill pattern		Monotonic		

Support and Multimaterial

- Support:
 - In Support you can change your Support and Filament for support settings. You can hover your cursor over the setting and read more about it
- Multimaterial:
 - In Multimaterial you can change the Prime tower and Flush options. You will NOT have to change these settings unless you print with multiple different filaments. You can hover your cursor over the setting and read more about it.

The image shows two screenshots of a printer's settings interface. The top screenshot is the 'Support' tab, and the bottom screenshot is the 'Multimaterial' tab.

Support Tab:

- Quality | Strength | **Support** | Multimaterial | Others
- Support**
 - Enable support: ☐
 - Type:
 - Threshold angle: °
 - Threshold overlap: mm or %
 - On build plate only: ☐
- Filament for Supports**
 - Support/raft base:
 - Support/raft interface:

Multimaterial Tab:

- Quality | Strength | Support | **Multimaterial** | Others
- Prime tower**
 - Enable: ☒
 - Width: mm
 - Prime volume: mm³
- Flush options**
 - Flush into objects' infill: ☐
 - Flush into objects' support: ☒